[Lecture Notebook] Evaluating Predictive Performance

MGS3701 Data Mining, Spring 2025

Chad (Chungil Chae)

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Table 5.1

```
# package forecast is required to evaluate performance
library(forecast)
```

```
Registered S3 method overwritten by 'quantmod':
  method            from
  as.zoo.data.frame zoo
```

```
Registered S3 methods overwritten by 'forecast':
  method  from
  head.ts stats
  tail.ts stats
```

```
# load file
toyota.corolla.df <- read.csv("data/ToyotaCorolla.csv")

# randomly generate training and validation sets
training <- sample(toyota.corolla.df$Id, 600)
validation <- sample(setdiff(toyota.corolla.df$Id, training), 400)

# run linear regression model
reg <- lm(Price~., data=toyota.corolla.df[,-c(1,2,8,11)], subset=training,
          na.action=na.exclude)
pred_t <- predict(reg, na.action=na.pass)
pred_v <- predict(reg, newdata=toyota.corolla.df[validation,-c(1,2,8,11)],
                  na.action=na.pass)

## evaluate performance
# training
accuracy(pred_t, toyota.corolla.df[training,]$Price)
```

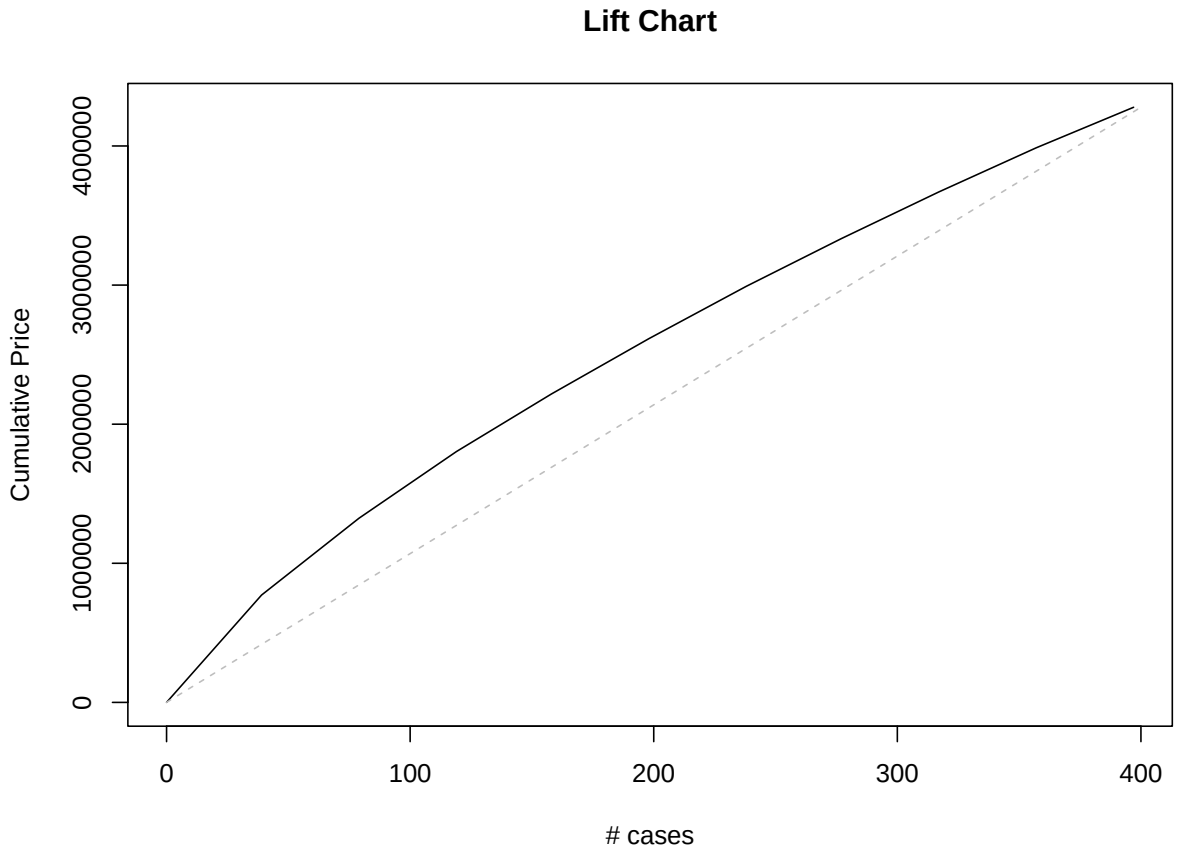
```
                     ME     RMSE      MAE        MPE     MAPE
Test set -7.06912e-11 1102.694 826.0141 -0.8096722 8.250023
```

```
# validation
accuracy(pred_v, toyota.corolla.df[validation,]$Price)
```

```
                ME     RMSE      MAE        MPE     MAPE
Test set 39.71589 1267.914 905.7535 -0.7389303 9.020261
```

Figure 5.2

```
# remove missing Price data
toyota.corolla.df <-
  toyota.corolla.df[!is.na(toyota.corolla.df[validation,]$Price),]

# generate random Training and Validation sets
training <- sample(toyota.corolla.df$Id, 600)
validation <- sample(toyota.corolla.df$Id, 400)

# regression model based on all numerical predictors
reg <- lm(Price~., data = toyota.corolla.df[,-c(1,2,8,11)], subset = training)

# predictions
pred_v <- predict(reg, newdata = toyota.corolla.df[validation,-c(1,2,8,11)])

# load package gains, compute gains (we will use package caret for categorical y later)
library(gains)
gain <- gains(toyota.corolla.df[validation,]$Price[!is.na(pred_v)], pred_v[!is.na(pred_v)])

# cumulative lift chart
options(scipen=999) # avoid scientific notation
# we will compute the gain relative to price
price <- toyota.corolla.df[validation,]$Price[!is.na(toyota.corolla.df[validation,]$Price)]
plot(c(0,gain$cume.pct.of.total*sum(price))~c(0,gain$cume.obs),
     xlab="# cases", ylab="Cumulative Price", main="Lift Chart", type="l")

# baseline
lines(c(0,sum(price))~c(0,dim(toyota.corolla.df[validation,])[1]), col="gray", lty=2)
```



18

```
# Decile-wise lift chart
barplot(gain$mean.resp/mean(price), names.arg = gain$depth,
        xlab = "Percentile", ylab = "Mean Response", main = "Decile-wise lift chart")
```

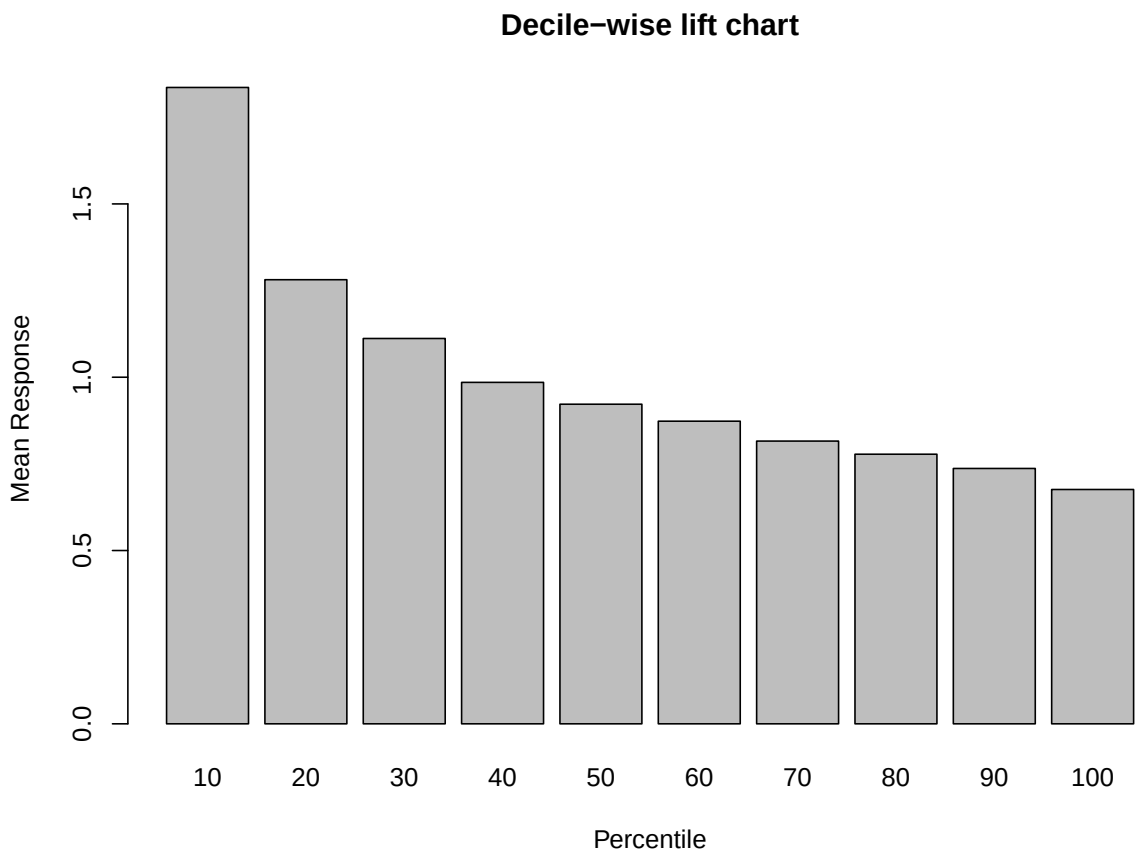


Table 5.5

```
library(caret)
```

```
Loading required package: lattice
```

```

Attaching package: 'caret'
```

```
The following object is masked from 'package:purrr':

    lift
```



```
library(e1071)

owner.df <- read.csv("data/ownerExample.csv")
#confusionMatrix(as.factor(ifelse(owner.df$Probability>0.5, 'owner', 'nonowner')),
#                owner.df$Class)
#confusionMatrix(as.factor(ifelse(owner.df$Probability>0.25, 'owner', 'nonowner')),
#                owner.df$Class)
#confusionMatrix(as.factor(ifelse(owner.df$Probability>0.75, 'owner', 'nonowner')),
#                owner.df$Class)
```

27 **Figure 5.4**

```
# replace data.frame with your own
df <- read.csv("data/liftExample.csv")


# create empty accuracy table
accT = c()

# compute accuracy per cutoff
for (cut in seq(0,1,0.1)){
  cm <- confusionMatrix(as.factor(1 * (df$prob > cut)), as.factor(df$actual))
  accT = c(accT, cm$overall[1])
}

# plot accuracy
plot(accT ~ seq(0,1,0.1), xlab = "Cutoff Value", ylab = "", type = "l", ylim = c(0, 1))
lines(1-accT ~ seq(0,1,0.1), type = "l", lty = 2)
legend("topright",  c("accuracy", "overall error"), lty = c(1, 2), merge = TRUE)
```

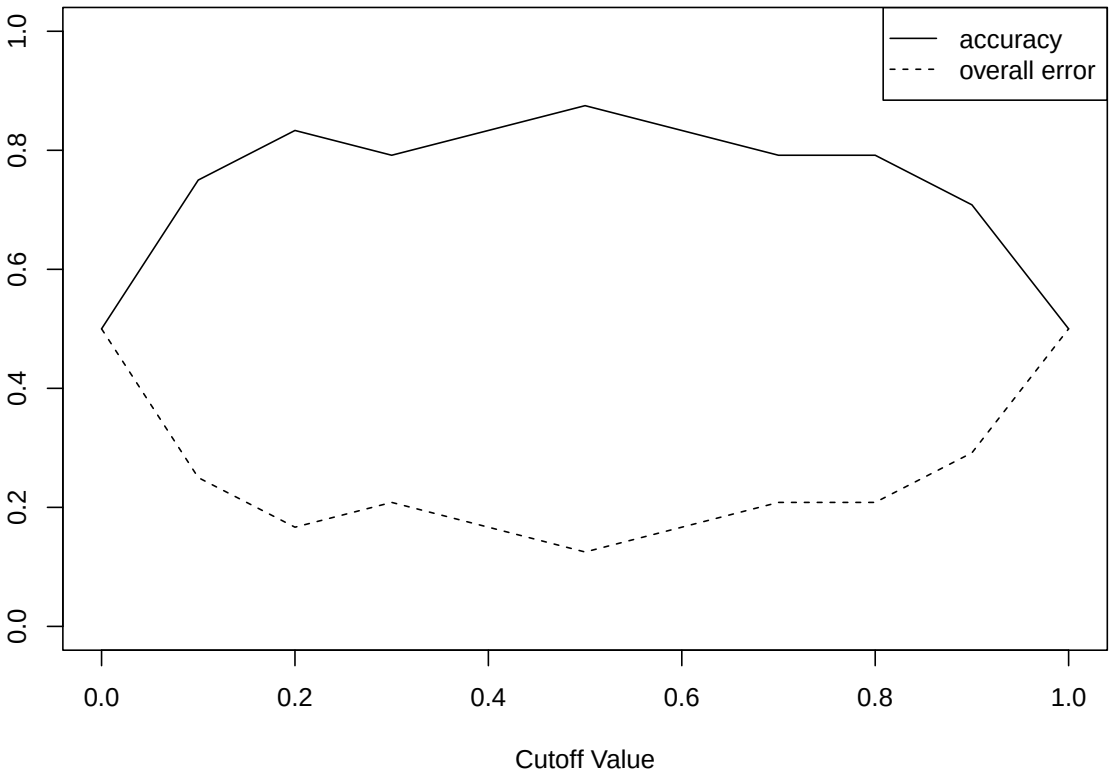


Figure 5.5

```
library(pROC)
```

```
Type 'citation("pROC")' for a citation.
```

```

Attaching package: 'pROC'
```

```
The following objects are masked from 'package:stats':

    cov, smooth, var
```

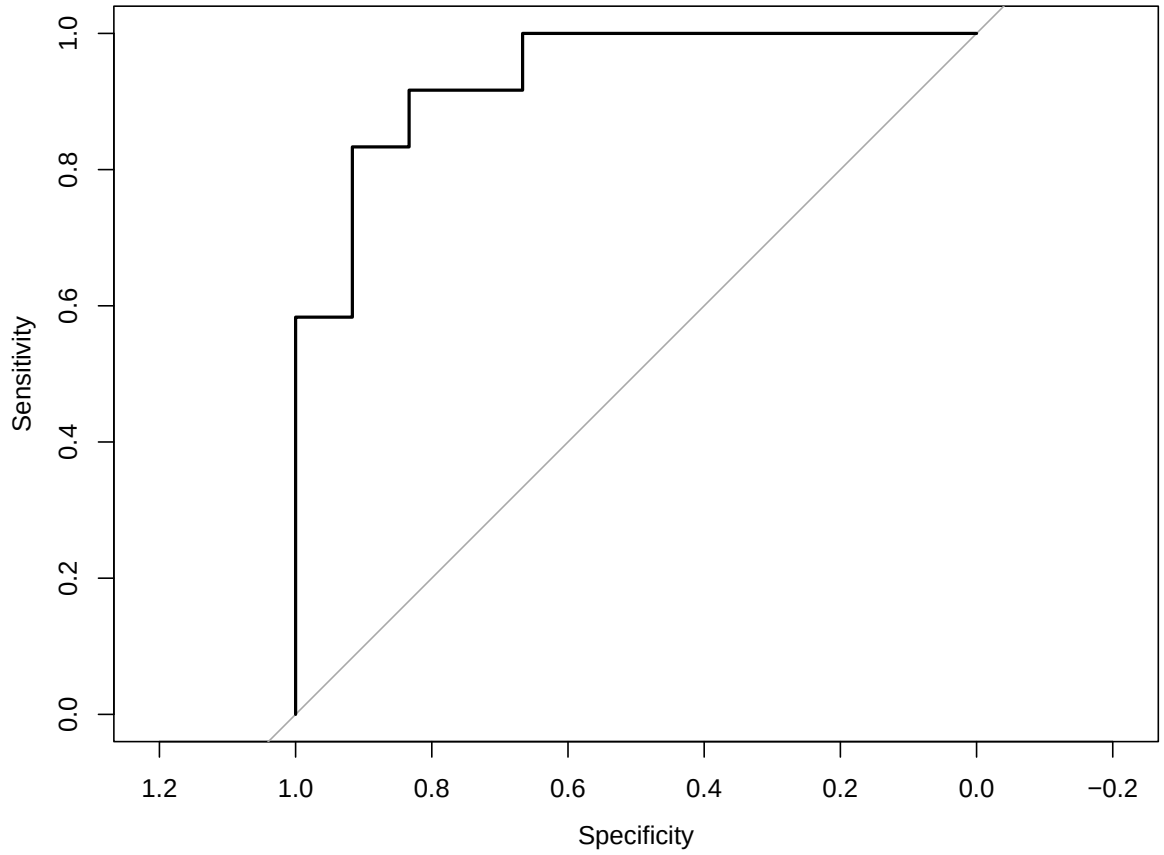
```
r <- roc(df$actual, df$prob)
```

```
Setting levels: control = 0, case = 1
```



```
Setting direction: controls < cases
```

```
plot.roc(r)
```

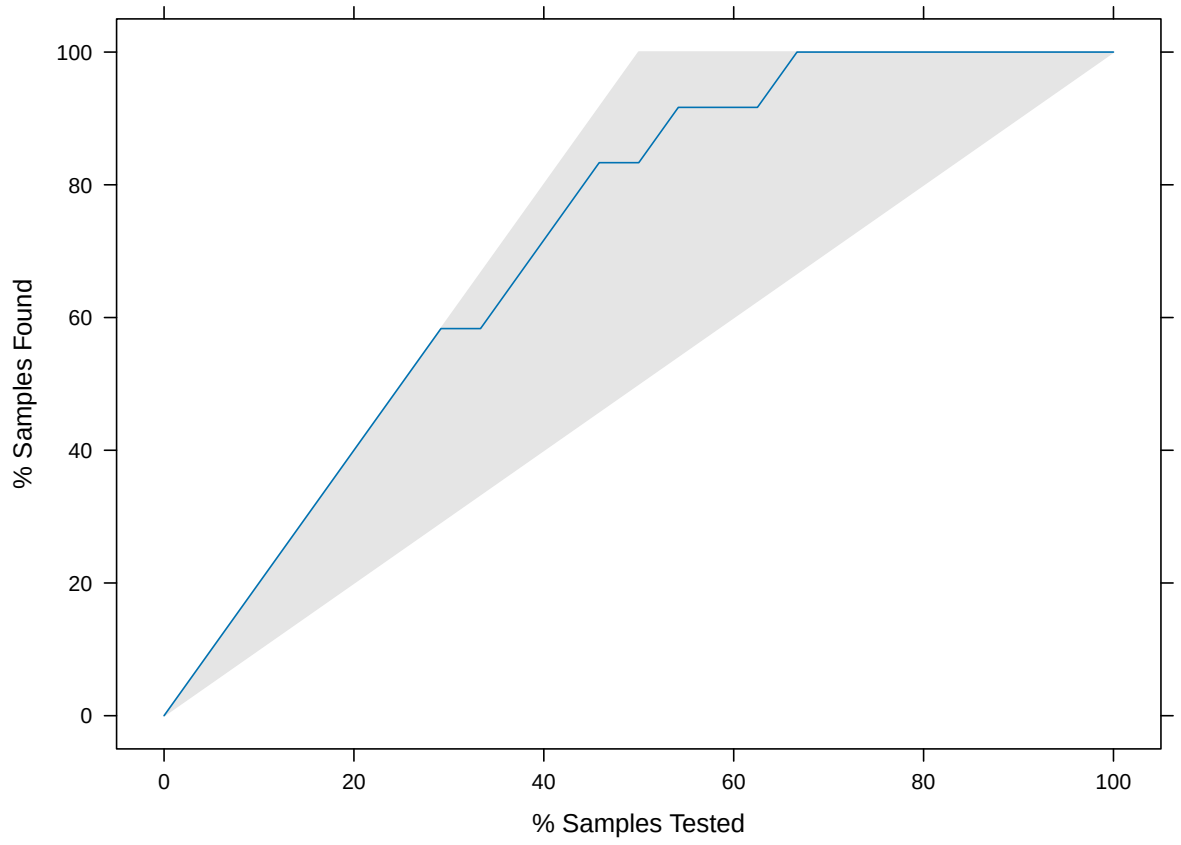



```
# compute auc
auc(r)
```

```
Area under the curve: 0.9375
```

Figure 5.6

```
# first option with 'caret' library:
library(caret)
lift.example <- lift(relevel(as.factor(actual), ref="1") ~ prob, data = df)
xyplot(lift.example, plot = "gain")
```



41

```
# Second option with 'gains' library:
library(gains)
df <- read.csv("data/liftExample.csv")
gain <- gains(df$actual, df$prob, groups=dim(df)[1])
plot(c(0, gain$cume.pct.of.total*sum(df$actual)) ~ c(0, gain$cume.obs),
     xlab = "# cases", ylab = "Cumulative", type="l")
lines(c(0,sum(df$actual))~c(0,dim(df)[1]), col="gray", lty=2)
```

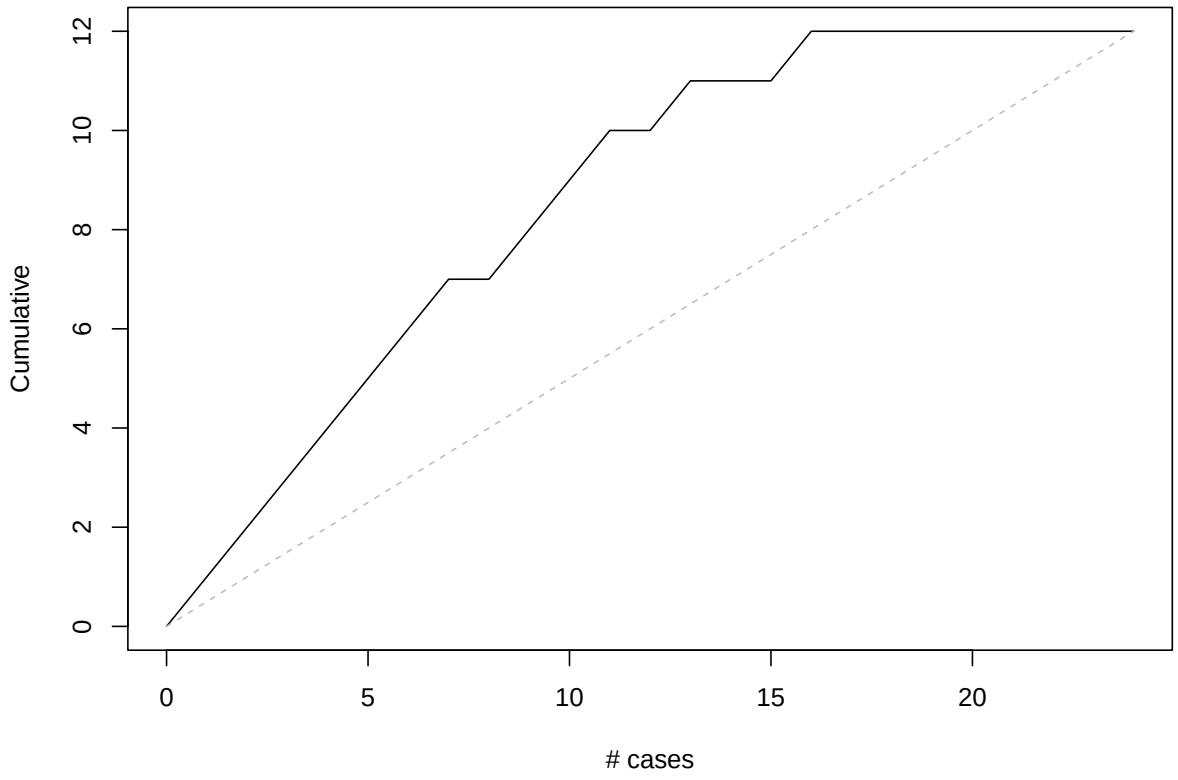
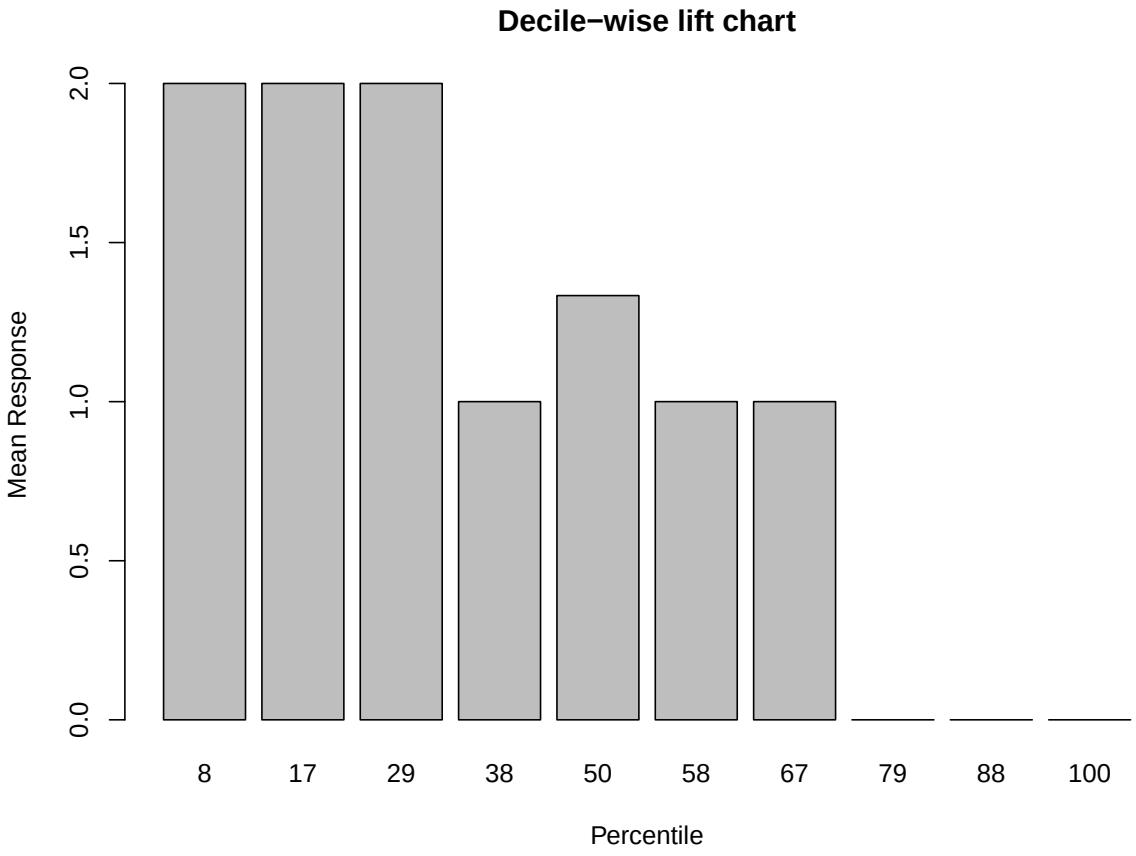


Figure 5.7

```
# use gains() to compute deciles.
# when using the caret package, deciles must be computed manually.

gain <- gains(df$actual, df$prob)
barplot(gain$mean.resp / mean(df$actual), names.arg = gain$depth, xlab = "Percentile",
        ylab = "Mean Response", main = "Decile-wise lift chart")
```

44

